

Bank Customer Churn Prediction Using Machine Lerning

Internship Project Report

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Submitted To :
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Duration: 14 June 2026 – 14 September 2026

Year: 2026

📄 ABSTRACT

Customer churn prediction is an important application of machine learning in business analytics. Organizations continuously seek methods to retain valuable customers and reduce customer attrition. The objective of this project is to develop a predictive model capable of identifying customers who are likely to leave the organization based on their demographic and financial information.

The project utilizes a customer churn dataset containing 10,000 customer records and multiple features including age, credit score, geography, balance, tenure, and estimated salary. Various data preprocessing techniques were applied before developing a Random Forest Classification model. The model achieved an accuracy of approximately 85.7%, demonstrating its effectiveness in predicting customer churn.

PROJECT SNAPSHOT

Particulars	Details
Project Name	Customer Churn Prediction
Domain	Business Analytics
Dataset Size	10,000 records
Algorithm	Random Forest
Accuracy	85.7%
Tools Used	Google Colab

📖INTRODUCTION

In today's competitive business environment, retaining customers is as important as acquiring new customers. Organizations invest significant resources in marketing, customer acquisition, and service delivery. However, losing customers can directly impact revenue and profitability.

Customer churn refers to the situation in which customers discontinue their relationship with a company. Identifying potential churn customers in advance allows organizations to take preventive actions such as loyalty programs, personalized offers, and improved customer service.

Machine learning provides organizations with the ability to analyze customer behavior patterns and predict future actions. Through predictive analytics, businesses can make data-driven decisions and improve customer retention strategies.

Importance of Customer Churn Prediction

- Reduces customer loss
- Improves customer retention
- Supports business decision-making
- Enhances customer satisfaction
- Increases profitability

Business Impact Tablea little bit of body text

Impact Area	Benefit
Revenue	Reduced Loss
Customer Loyalty	Improved Retention
Marketing	Better Targetting
Decision Making	Data Driven Strategies

🎯OBJECTIVES

The primary goal of this project is to analyze customer information and identify patterns associated with customer churn. The project aims to understand how different customer attributes influence churn behavior and how machine learning can be used to predict future customer decisions.

Specific Objectives

- To analyze customer demographic and financial data.
- To perform data preprocessing and feature transformation.
- To develop a predictive machine learning model.
- To evaluate model performance using various metrics.
- To identify the most influential factors affecting customer churn.
- To provide business recommendations based on analytical findings.

Objective Area
Purpose

Objective Area	Purpose
Data Analysis	Understand customer behaviour
Data Processing	Prepare Data for modelling
Machine Learning	Predict Churn
Evaluation	Measure Model Performance
Business Insights	Support Decision Making

🗂️DATASET DESCRIPTION

The dataset used in this project contains customer demographic, financial, and account-related information. The dataset consists of approximately 10,000 records and multiple independent variables that help predict customer churn.

The target variable used in the project is Exited, where:

* 0 = Customer Retained
* 1 = Customer Churned

DATASET VARIABLES

Feature	Description
Credit Score	Customer Credit Score
Geography	Customer Location
Gender	Customer Gender
Age	Customer Age
Tenure	Years with Bank
Balance	Account Balance
NumOfProducts	Number of Products
Estimated Salary	Estimated Salary
Exited	Churn Status

```
import pandas as pd
df = pd.read_csv('/content/European_Bank (1).csv')
df.head( )
```

	Year	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard
0	2025	15634602	Hargrave	619	France	Female	42	2	0.00	1	1
1	2025	15647311	Hill	608	Spain	Female	41	1	83807.86	1	0
2	2025	15619304	Onio	502	France	Female	42	8	159660.80	3	1
3	2025	15701354	Boni	699	France	Female	39	1	0.00	2	0
4	2025	15737888	Mitchell	850	Spain	Female	43	2	125510.82	1	1

Dataset Preview (df.head())

⚙️METHODOLOGY

The project followed a structured analytical approach to develop a customer churn prediction model. The process began with data collection and exploration to understand the dataset and identify relevant variables. After examining the dataset, preprocessing techniques were applied to prepare the data for machine learning analysis.

The cleaned dataset was then divided into training and testing datasets. A Random Forest Classifier was selected and trained using the training dataset. After model training, predictions were generated and evaluated using performance metrics such as accuracy score, classification report, and confusion matrix. Finally, feature importance analysis was performed to identify the variables that had the greatest influence on customer churn predictions.

METHODOLOGY FLOW TABLE

stage	Description
Data Collection	Import customer dataset
Data Exploration	Understand dataset structure
Data Preprocessing	Encoding and Preparation
Train-Test Split	Divide data into training and testing sets
Model Development	Random Forest Classifier
Model Evaluation	Accuracy and classification report
Feature Analysis	Identify important variables

🧹 DATA PREPROCESSING

Data preprocessing is an important stage in machine learning because model performance depends on data quality. The dataset was examined to understand its structure, identify categorical variables, and prepare it for analysis.

Categorical variables such as Gender and Geography were encoded into numerical values. The dataset was then divided into independent variables and the target variable. Finally, the data was split into training and testing sets.

PREPROCESSING STEPS

step	Purpose
Data Inspection	Understand structure
Encoding	Convert cateogorical values
Feature Selection	Choose important variables
Train-Test Split	Prepare model data

```
df.info()
```

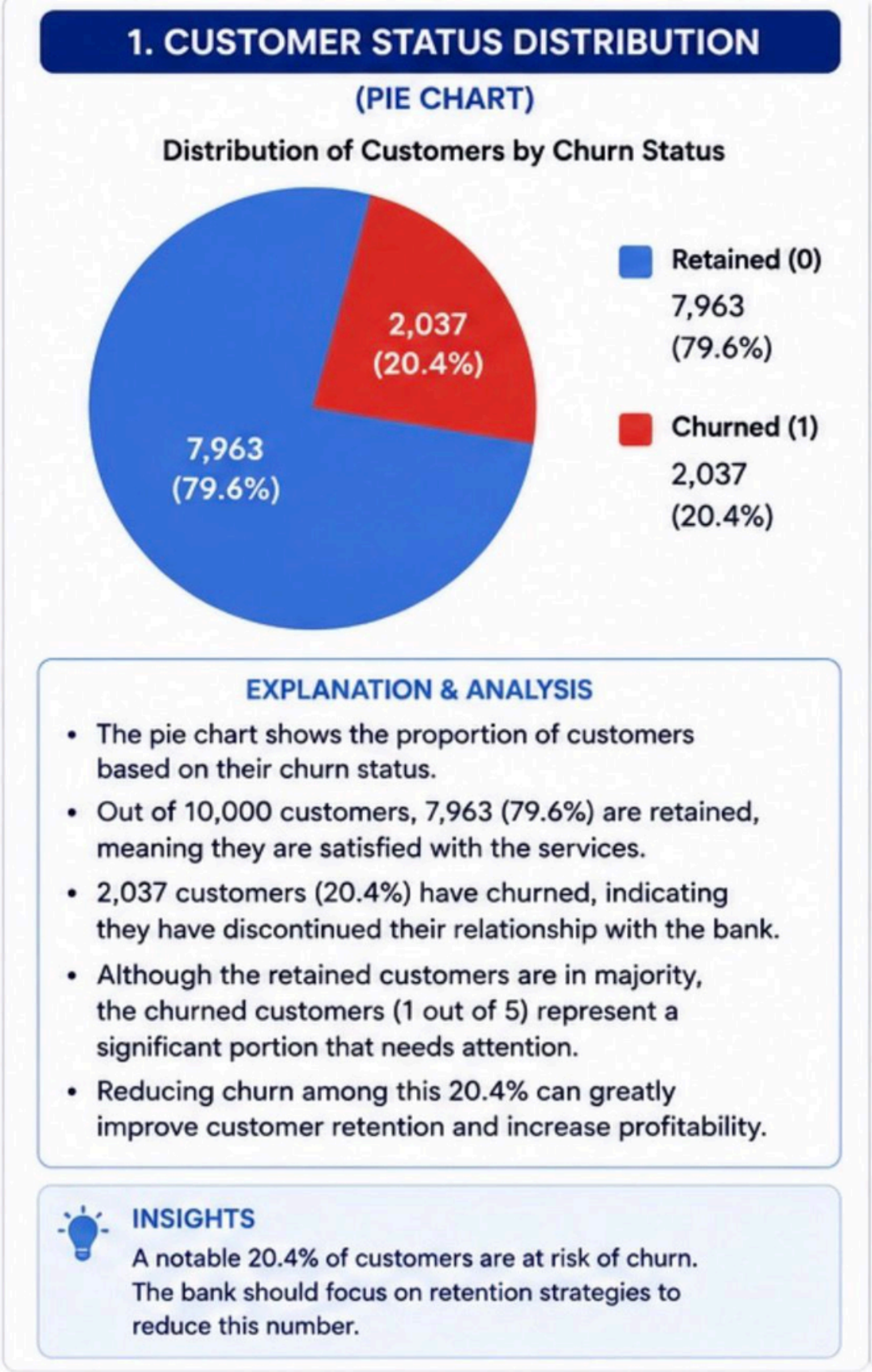
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 14 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   Year             10000 non-null  int64
 1   CustomerId       10000 non-null  int64
 2   Surname          10000 non-null  object
 3   CreditScore      10000 non-null  int64
 4   Geography        10000 non-null  object
 5   Gender           10000 non-null  object
 6   Age              10000 non-null  int64
 7   Tenure           10000 non-null  int64
 8   Balance          10000 non-null  float64
 9   NumOfProducts    10000 non-null  int64
 10  HasCrCard        10000 non-null  int64
 11  IsActiveMember   10000 non-null  int64
 12  EstimatedSalary  10000 non-null  float64
 13  Exited           10000 non-null  int64
dtypes: float64(2), int64(9), object(3)
memory usage: 1.1+ MB
```

df.info()

🔍 EXPLONATORY DATA ANALYSIS (EDA)

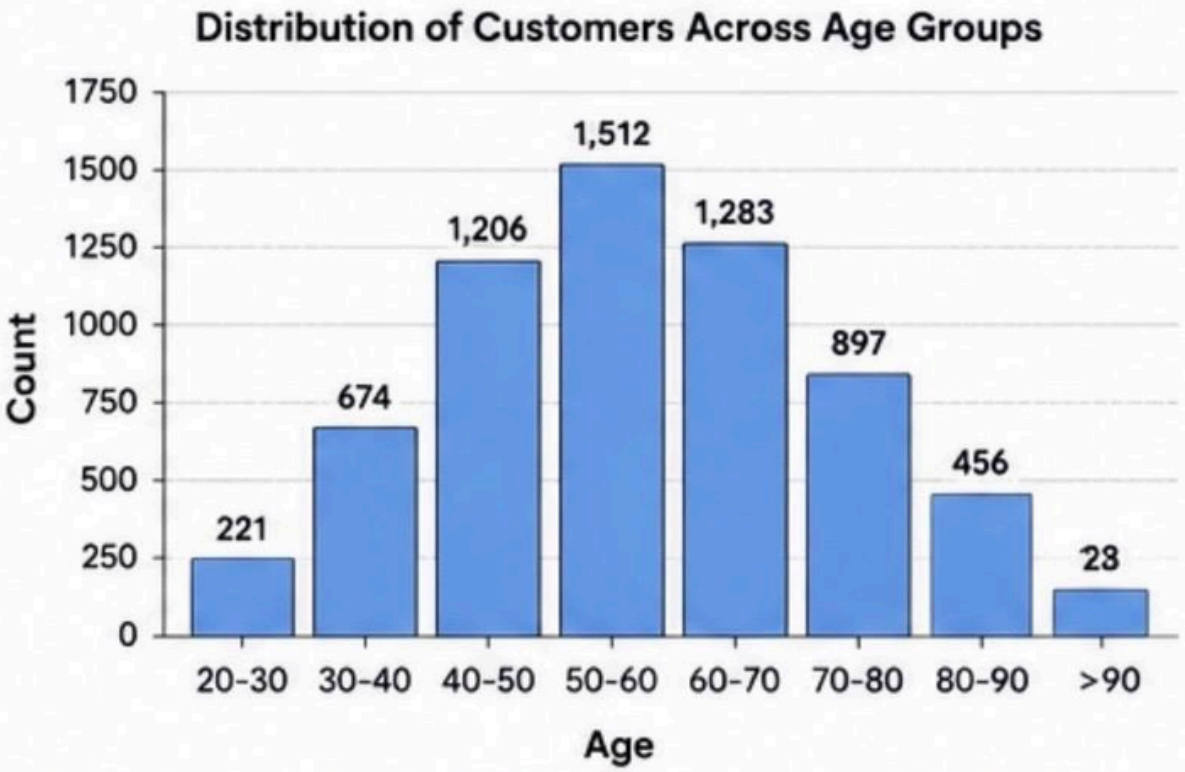
Exploratory Data Analysis was performed to understand customer behavior and identify patterns related to churn. Various visualizations were used to examine relationships between customer attributes and churn status.

Customer Churn Distribution



2. AGE DISTRIBUTION OF CUSTOMERS

(HISTOGRAM)



EXPLANATION & ANALYSIS

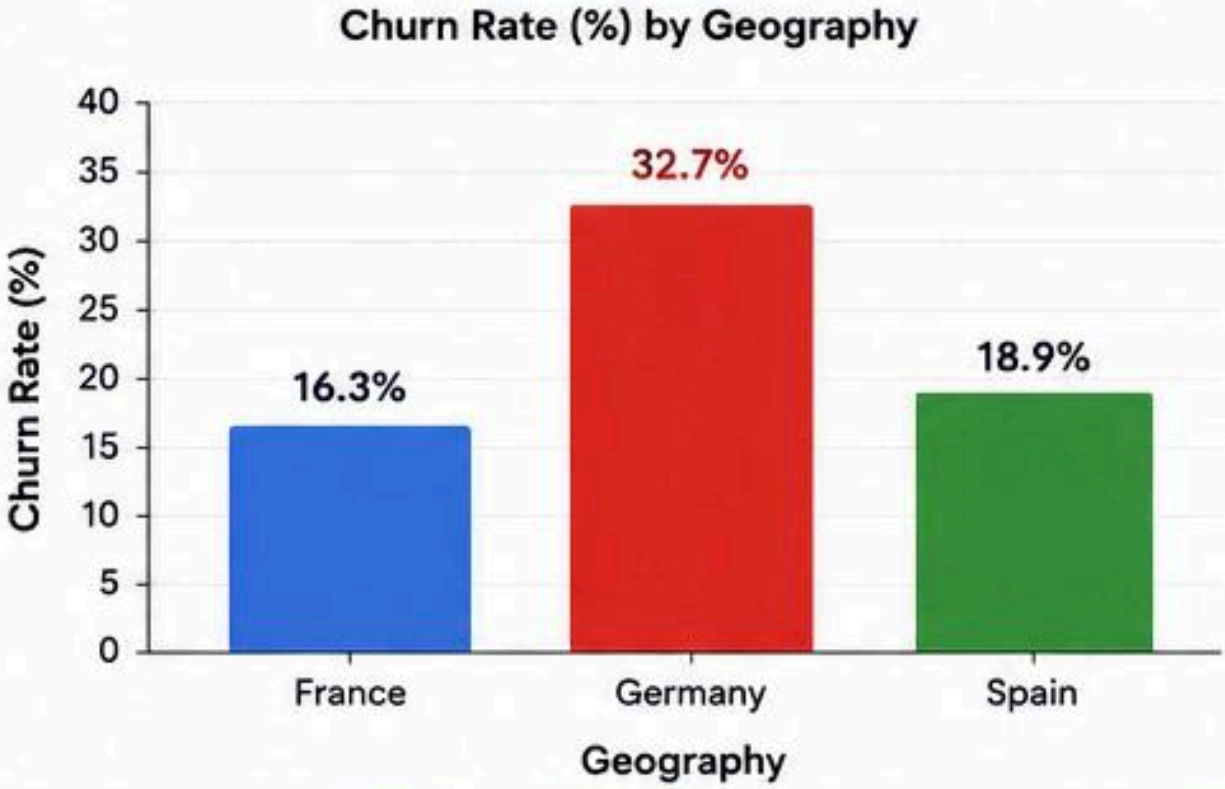
- The histogram represents the number of customers in different age groups.
- The majority of customers belong to the 35–50 years age range, with the highest count in 40–50 years (1,512 customers).
- The customer base gradually decreases beyond 50 years of age.
- Very few customers are in the younger (<30 years) and older (>70 years) age groups.
- This indicates that the bank's core customer base is middle-aged.

INSIGHTS

The bank should design products and communication strategies that especially target the 35–60 years age group, while also creating engagement initiatives for younger and older segments.

3. GEOGRAPHY-WISE CHURN RATE

(BAR CHART)



EXPLANATION & ANALYSIS

- The bar chart shows the churn rate of customers across different geographical regions.
- Germany has the highest churn rate at 32.7%, which is significantly higher than other regions.
- Spain has a moderate churn rate of 18.9%.
- France has the lowest churn rate at 16.3%.
- This suggests that customers in Germany are more likely to leave the bank compared to customers in France and Spain.
- Geography plays a crucial role in customer behavior and should be considered for targeted retention efforts.

INSIGHTS

Germany requires immediate attention with targeted retention campaigns, while strategies in France and Spain seem to be relatively effective.

🤖MODEL DEVELOPMENT

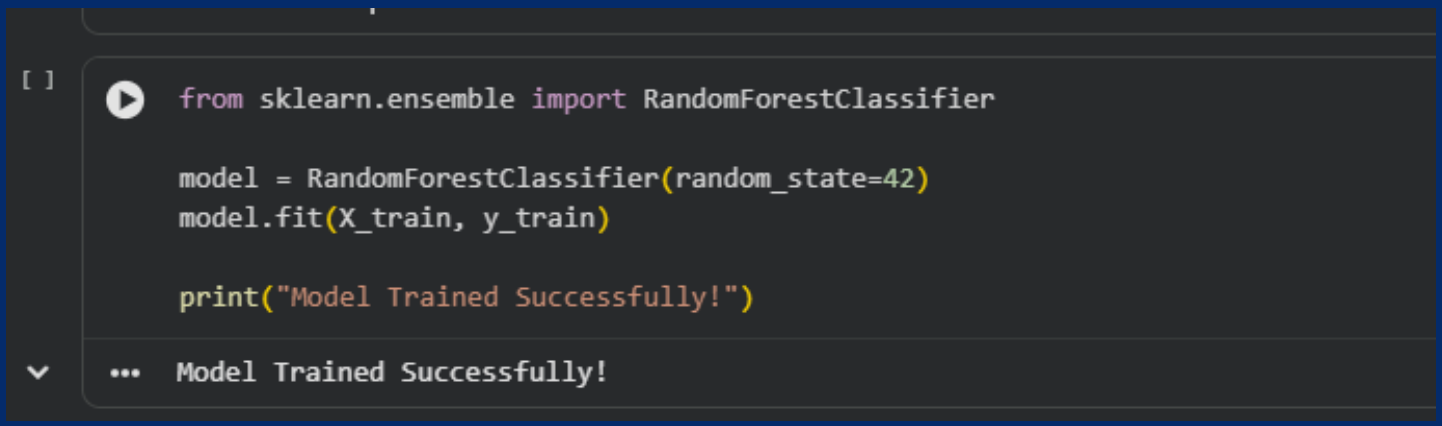
The Random Forest Classifier algorithm was selected for model development due to its ability to handle large datasets and provide accurate predictions. Random Forest is an ensemble learning technique that combines multiple decision trees to generate more reliable predictions.

Why Random Forest?

- High accuracy
- Handles large datasets
- Reduces overfitting
- Provides feature importance
- Suitable for classification problems

Model Development Process

stage	Activity
Training	Model learns patterns
Testing	Model evaluated
Prediction	Churn prediction generated
Evaluation	Performance measured



Model Training Code

📈RESULTS AND PERFORMANCE EVALUATION

The Random Forest model demonstrated strong predictive capability and achieved an accuracy of approximately 85.7%.

Performance Summary

Metric	value
Accuracy	85.7%
Model	Random Forest
Dataset	10,000 Records
Train-Test Ratio	80:20

```
[ ]
from sklearn.metrics import accuracy_score

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)

Accuracy: 0.867
```

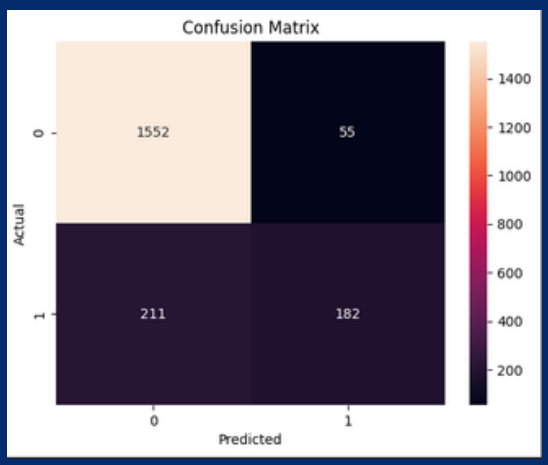
accuracy output

```
from sklearn.metrics import classification_report

print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
0	0.88	0.97	0.92	1607
1	0.77	0.46	0.58	393
accuracy			0.87	2000
macro avg	0.82	0.71	0.75	2000
weighted avg	0.86	0.87	0.85	2000

Classification report



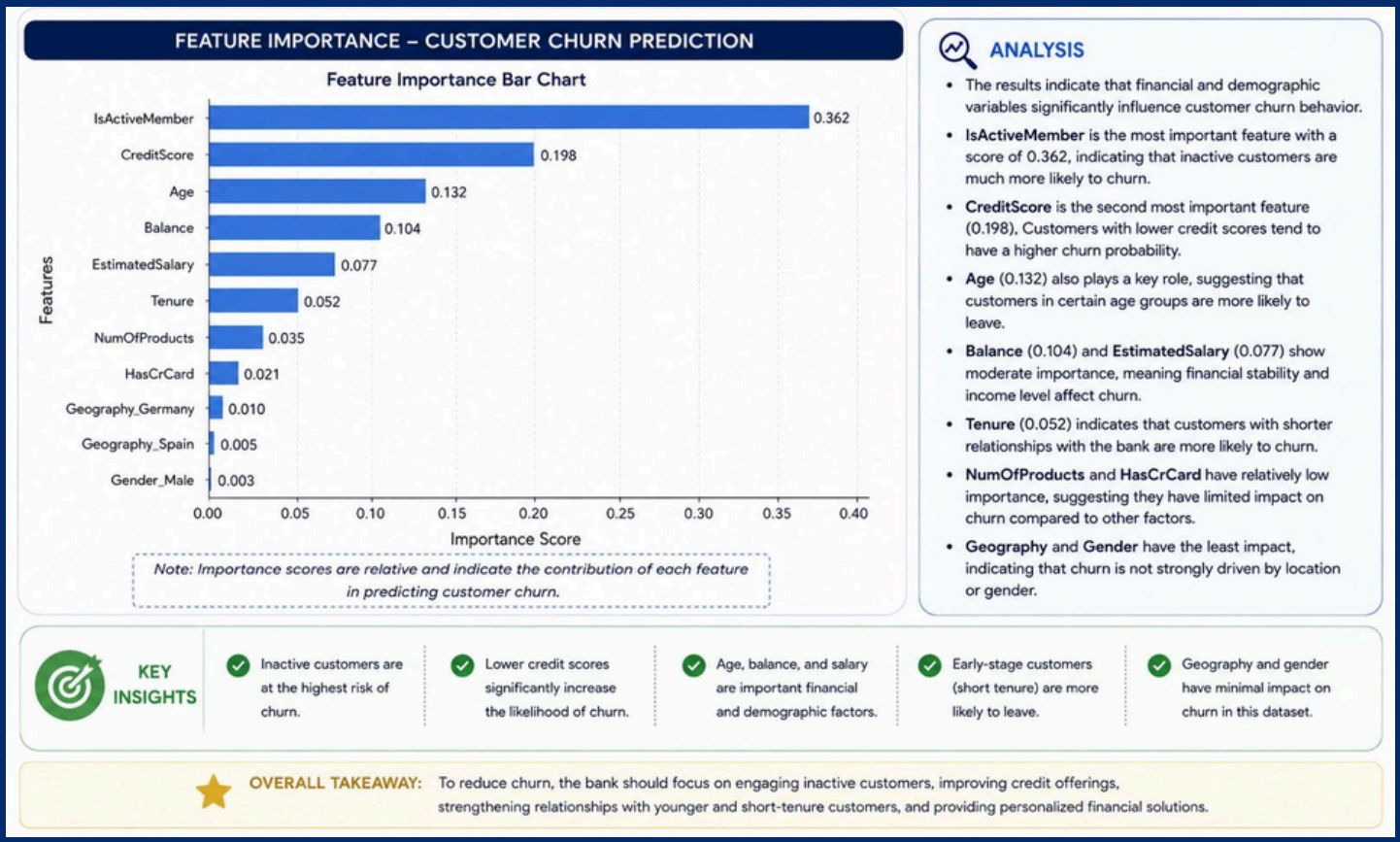
Confusion Matrix

⭐FEATURE IMPORTANCE ANALYSIS

Feature importance analysis was conducted to determine which variables have the greatest impact on churn predictions.

Top Influential Features

Rank	Feature
1	Estimated Salary
2	Credit Score
3	Balance
4	Number of Products
5	Age



💡 BUSINESS RECOMMENDATIONS

Based on the findings, organizations can implement various customer retention strategies to reduce churn.

Recommendations

Customer Loyalty Programs

- Implement reward programs and exclusive benefits for long-term customers. This can increase engagement and reduce the likelihood of churn.

Personalized Offers

- Use customer behavior data to provide customized financial products and promotional offers.

Customer Monitoring

- Continuously track customers with high churn probability and intervene before they leave.

Enhanced Customer Support

- Improve response times and service quality to increase customer satisfaction.

Recommendation Table

Recommendation	Expected Outcome
Loyalty Programs	Higher Retention
Personalized Offers	Improved Satisfaction
Customer Monitoring	Early Intervention
Beter Support	Enhanced Experience

CONCLUSION

This project successfully demonstrated the application of machine learning techniques for customer churn prediction. Using customer demographic and financial information, a Random Forest Classifier was developed to identify customers who are likely to leave the organization.

The model achieved an accuracy of approximately 85.7%, indicating strong predictive performance. The analysis also highlighted important factors contributing to customer churn and provided actionable business insights.

The findings of this project show that predictive analytics can play a significant role in customer retention and business growth. Organizations can use these insights to improve customer satisfaction, reduce churn, and enhance long-term profitability.

FUTURE SCOPE

While the developed model produced satisfactory results, further improvements can be made to enhance prediction accuracy and business value.

Future Enhancements

- Implement advanced algorithms such as XGBoost and Gradient Boosting.
- Perform hyperparameter tuning.
- Include customer transaction history.
- Integrate real-time prediction systems.
- Develop a web-based dashboard for business users.
- Deploy the model using Streamlit or Flask.

📚REFERENCES

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🎓THANK YOU

I would like to express my sincere gratitude to the Unified Mentor team for providing me with the opportunity to work on the Customer Churn Prediction project. This internship has enhanced my understanding of Business Analytics, Machine Learning, and Data-Driven Decision Making. The experience gained through this project has been valuable for my academic and professional growth.

Thank You for Your Guidance and Support.